

Data Structures and Algorithms

1. Introduction

This section introduces the book and its contents, covering what the book is about, the assumed knowledge required, and how to work through the examples. It provides an overview of the book's structure and testing.

2. Linked Lists

A linked list is a data structure where each element points to the next one, allowing for efficient insertion and deletion. It's a fundamental concept in programming, and understanding it is crucial for working with more complex data structures.

3. Binary Search Tree

A binary search tree is a data structure where each node has at most two children, and the left child is less than the parent, while the right child is greater. It's used for efficient searching, insertion, and deletion of elements.

4. Heap

A heap is a specialized data structure that satisfies the heap property, where the parent node is either greater than (max heap) or less than (min heap) its child nodes. It's used in priority queues and graph algorithms.

5. Sets

A set is an unordered collection of unique elements, and it can be used to store and manipulate data. Sets can be ordered or unordered, and they have various operations like insertion, deletion, and union.

6. Queues

A queue is a data structure that follows the First-In-First-Out (FIFO) principle, where elements are added and removed from the end and front respectively. It's used in job scheduling, print queues, and other applications.

7. AVL Tree

An AVL tree is a self-balancing binary search tree that ensures the height of the tree remains relatively small by rotating nodes when the balance factor becomes too large. It's used for efficient searching, insertion, and deletion.

8. Sorting

Sorting algorithms arrange data in a specific order, either ascending or descending. Various sorting algorithms like Bubble Sort, Merge Sort, and Quick Sort have different time complexities and use cases.

9. Numeric Algorithms

Numeric algorithms perform operations like primality testing, base conversions, and calculating the greatest common denominator. They're essential in various fields like cryptography, coding theory, and numerical analysis.

10. Searching

Searching algorithms find a specific element or pattern within a data structure. Different searching algorithms like Sequential Search and Probability Search have varying time complexities and applications.